



Fig. 2: Pin configurations of IRF540, SK100 and TL431

the combination of resistor R6 and diode D2 to turn on transistor T2, which is an n-channel power MOSFET. As a result, the DC supply from input pin J1 is routed to the output load connected at socket J2.

However, if the battery voltage drops below 10V, the reference terminal voltage of IC1 falls below 2.5V and its cathode goes high. Transistor T1 unbiases to turn off transistor T2 and the output load disconnects to prevent deep discharge of the storage bat-



Fig. 3: Cigarette lighter socket extension cable

tery. The red LED (LED1) is used as a simple output-power status indicator. SPST toggle switch S1 is the power-on/off-cum-reset switch.

For testing the circuit, apply 12V DC to input point J1 and adjust trimmer VR1 (22-kilo-ohm) such that LED1 lights up and the output

voltage is available at J2. Recheck the calibration to ensure that the output supply is disabled when the input voltage falls to a value of about 10V DC.

IC1 is a three-terminal adjustable shunt regulator. Its output voltage may be set at any level greater than 2.5V (V_{Ref}) and up to 3.6V by selecting resistors R1 and R2 that act as a voltage-divider network. T2 is a general-purpose n-channel power MOSFET. Any MOSFET with similar characteristics can also be used in place of T2. Use a heat sink for T2 to allow heat dissipation.

After construction, enclose the circuit in a small metallic or plastic box with holes to mount the power switch (S1) and indicator (LED1). Output jack J2 should match with the appliance to be used. ●

1-30 Minute Timer

S.C. DWIVEDI

Using this circuit you can switch on an appliance for a desired time. The circuit provides selectable time settings of 1, 2, 5, 10, 15 and 30 minutes, and can be used for domestic as well as industrial applications.

The circuit can be divided into two sections—power supply and timer. The power supply is built around transformer X1, bridge recti-

fier BR1, capacitor C1 and 12V voltage regulator IC LM7812 (IC1). The 230V AC mains supply is stepped down by transformer X1 to deliver the secondary output of 12V, 250 mA. The transformer output is rectified by full-wave bridge rectifier BR1, filtered by capacitor C1 and regulated by IC1. The circuit can also be powered by a 12V battery. The power source can be selected by using switch S2.

The timer section is built around



Selection of the Output Time at pin 3 of Timer IC2

Position of rotary switch 1	Time (approx.) in minutes at pin 3 of timer IC NE555
Position 1	1
Position 2	2
Position 3	5
Position 4	10
Position 5	15
Position 6	30

